# SOFTWARE TEST PLAN (STP)

## ONLINE QUIZ SYSTEM

Version 1.0 | 12 October 2026

| Document information | Details |
|---|---|
| **Project** | Online Quiz System |
| **Document** | Software Test Plan (STP) |
| **Course / Department** | Software Engineering |
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| **Faculty guide** | Arpitha K |
| **Baseline documents** | Online Quiz System SRS v1.0 (12 Oct 2026); Online Quiz System SAD v1.0 |
| **Status** | Submission draft |

Purpose: Define how the Online Quiz System will be verified against every functional, non-functional, security and CI/CD requirement in SRS v1.0, including scope, strategy, environment, schedule, responsibilities, test cases and traceability.

# Document control

## Revision history

| Version | Date | Description |
|---|---|---|
| 1.0 | 12 Oct 2026 | Initial test plan aligned with SRS v1.0 |

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Document convention: TC = test case; AT = acceptance test defined in SRS Section 6; FR, NFR, SEC and SO identifiers refer to SRS v1.0. Levels: U = unit, I = integration, S = system, A = acceptance, P = performance. Priority: H = high, M = medium, L = low.

# 1. Introduction

## 1.1 Purpose

This Software Test Plan defines the objectives, scope, strategy, environment, schedule, responsibilities, test cases and traceability for verifying the Online Quiz System v1.0. It shows how every requirement in the SRS will be verified and what evidence is required before the system is accepted.

**Status of testing:** this is a plan. All test cases, tools and activities described here are planned; none has been executed at the time of writing. Actual results will be recorded in the test execution log and the Test Summary Report during the November 2026 test window.

## 1.2 Scope

Testing covers the complete browser-based system described in SRS v1.0: registration, login and role-based access; quiz discovery; quiz authoring, publication and validation; timed quiz attempts with server-enforced expiry and single-attempt rules; automatic scoring; score history; per-quiz leaderboards; feedback and administration; responsive UI; security controls; and the GitHub Actions CI pipeline.

Out of scope (consistent with SRS Section 1.2): paid examinations, live video proctoring, question-generation AI, offline quiz attempts and native mobile apps. See Section 4 for the full list of exclusions.

## 1.3 Objectives

- Verify that all 31 functional, 7 non-functional and 6 security requirements, and both security objectives, in SRS v1.0 are met.

- Confirm that acceptance tests AT-01 to AT-10 pass before submission and demonstration.
- Protect quiz fairness: one attempt per quiz, server-enforced deadlines, hidden answer keys and correct, deterministic ranking.
- Keep the main branch releasable through automated tests, at least 80% statement coverage and a passing CI run.
- Find, report and track defects to closure, and publish a Test Summary Report.

### 1.4 References

| Reference | Description |
|---|---|
| **Online Quiz System SRS v1.0 (revised)** | Requirements baseline (FR-001 to FR-031, NFR-001 to NFR-007, SEC-001 to SEC-006, SO-01, SO-02, AT-01 to AT-10) |
| **Online Quiz System SAD v1.0** | Architecture, components, API design and sequence diagrams used to design integration tests |
| **IEEE 829 / ISO/IEC/IEEE 29119-3** | Standard structure for software test documentation |
| **OWASP Top 10 and OWASP Web Security Testing Guide** | Basis for the security validation in Section 5.3 |
| **Test Plan Template for SE** | Course template on which this document is based |
| **Complete CI/CD Project Instructions** | Course guidelines for the CI pipeline stages, quality gates and pipeline artifacts |

### 1.5 Definitions and acronyms

| Term | Meaning |
|---|---|
| **STP** | Software Test Plan (this document) |
| **SRS / SAD** | Software Requirements Specification / Software Architecture and Design document |
| **TC** | Test case, identified as TC-<MODULE>-<NN> |
| **AT** | Acceptance test defined in SRS Section 6 |
| **RTM** | Requirements Traceability Matrix |
| **UAT** | User acceptance testing (final demonstration to the faculty guide) |
| **p95** | 95th-percentile response time: 95% of requests complete within this time |
| **Smoke test** | Short check of critical flows (login, start quiz, submit, view score) on a new build |
| **Regression test** | Re-running existing tests after a change to confirm nothing else broke |
| **JWT / RBAC / CI** | JSON Web Token / Role-Based Access Control / Continuous Integration |
| **Attempt** | One student submission for a specific quiz (as defined in the SRS) |

## 2. Test Items

The item under test is the Online Quiz System v1.0: the React/Vite client and the Node.js/Express API with MongoDB persistence, built from the main branch. The commit SHA of every build that is tested is recorded in the execution log and the Test Summary Report.

| Item | Test item / module | Description | SRS reference |
|---|---|---|---|
| **TI-01** | Authentication and access control | Register, login, logout, JWT issuing/validation, role middleware | FR-001 to FR-004, SEC-001, SEC-002 |
| **TI-02** | Quiz discovery and site pages | Quiz Topics, quiz details, Home, About, Contact, Feedback pages | FR-005 to FR-007 |
| **TI-03** | Quiz management | Create, edit, delete, publish and unpublish quizzes and questions | FR-008 to FR-012 |
| **TI-04** | Quiz attempt engine | Start attempt, save answers, countdown, submit, auto-finalize on expiry | FR-013 to FR-018, NFR-002, NFR-003 |
| **TI-05** | Scoring, history and leaderboard | Score calculation, result page, My Scores, per-quiz leaderboard | FR-019 to FR-023, SEC-003 |
| **TI-06** | Feedback and administration | Feedback submission and review, user role management | FR-024 to FR-026 |
| **TI-07** | REST API and data layer | Express routes, input validation, Mongoose models, error handling | SEC-004, NFR-001 |
| **TI-08** | Frontend UI | Responsive layouts, forms, labels and validation messages | NFR-004, NFR-005 |
| **TI-09** | Repository and CI pipeline | Branching/PR workflow, GitHub Actions workflow, README and API docs | FR-027 to FR-031, NFR-006, NFR-007, SEC-005, SEC-006 |

# 3. Features to be Tested

All requirements in SRS v1.0 are in scope. The table groups them by feature area, matching the SRS RTM groups. The per-requirement mapping to test cases is in Section 13.3.

| Feature area | SRS requirement IDs | Test focus | Priority |
|---|---|---|---|
| **Authentication and access** | FR-001 to FR-004 | Registration, login, logout, role enforcement on the server | High |
| **Navigation and discovery** | FR-005 to FR-007 | Only published quizzes shown; quiz metadata; page reachability | Medium |
| **Quiz management** | FR-008 to FR-012 | Draft creation, question validation, ownership, publish rules | High |
| **Quiz attempts and timeout** | FR-013 to FR-018 | Single attempt, answer saving, submit, server-side expiry, late updates | High |
| **Scoring and leaderboards** | FR-019 to FR-023 | Correct scores, hidden answer keys, history, per-quiz ranking and tie-break | High |
| **Feedback and administration** | FR-024 to FR-026 | Feedback submission and review, protected role management | Medium |
| **Version control and CI** | FR-027 to FR-031 | Triggers, failing/passing pipelines, artifacts, coverage gate, audit | High |
| **Performance and reliability** | NFR-001 to NFR-003 | p95 under load, idempotent submission, server-time expiry | High |
| **Usability and maintainability** | NFR-004 to NFR-007 | 375 px layouts, form errors, README/API docs, 80% coverage | Medium |
| **Security** | SEC-001 to SEC-006, SO-01, SO-02 | Hashing, JWT/RBAC, answer-key exposure, input validation, secrets, audit | High |

# 4. Features Not to be Tested

| Feature / item | Reason |
|---|---|
| **Paid examinations, live video proctoring, AI question generation, offline attempts, native mobile apps** | Out of scope in SRS Section 1.2; not implemented |
| **Internals of MongoDB Atlas, GitHub Actions runners, browsers and npm packages** | Third-party platforms; only our integration with them is tested |
| **Production hosting / deployment** | Deployment is optional in SRS Appendix B; tests run on local and CI environments |
| **Load beyond 50 concurrent users, stress and soak testing** | NFR-001 defines 50 concurrent users as the target; higher loads are not required |
| **Full penetration test by an external team** | Outside the project's time and resources; Section 5.3 covers targeted security tests |
| **Legacy browsers (e.g. Internet Explorer)** | SRS operating environment specifies modern desktop/mobile browsers |

# 5. Test Approach / Strategy

Testing is risk-based and automation-first. Logic that decides fairness (attempt rules, deadlines, scoring, ranking and access control) is covered by automated unit and API tests that run in CI on every push and pull request. System and acceptance tests confirm complete user flows in the browser. Test cases are written from the SRS acceptance criteria before or alongside implementation, so each feature branch contains its own tests.

## 5.1 Test levels: unit, integration, system and acceptance

| Level | What is tested | Tools | Performed by / where |
|---|---|---|---|
| **Unit (U)** | Scoring function, tie-break comparator, deadline/expiry logic, validators, React components and form behaviour | Vitest, React Testing Library | Developer of the feature; locally and in CI |
| **Integration (I)** | Express routes with auth/role middleware, validation and MongoDB test database; API response contents | Supertest, Vitest, mongodb-memory-server or local MongoDB test DB | Developers / QA lead; locally and in CI |
| **System (S)** | End-to-end flows in the browser on a full build with seeded data: register to leaderboard, authoring to publication, CI behaviour | Browsers, Chrome DevTools, Postman, GitHub Actions | QA lead and team; test environment |
| **Acceptance (A)** | SRS acceptance tests AT-01 to AT-10 demonstrated on the release candidate | Manual demonstration and CI run evidence | Whole team with the faculty guide |

## 5.2 Test types: functional, regression, performance and usability

| Type | Approach | Requirements |
|---|---|---|
| **Functional** | Positive, negative and boundary tests derived from each SRS acceptance criterion (Section 13.2) | FR-001 to FR-031 |
| **Regression** | Full automated suite runs in CI on every push and pull request; smoke test on every new build; failed TCs re-run after each fix | All automated TCs |
| **Performance** | Planned k6 load script with 50 virtual users for 5 minutes on the main API flows; concurrent-submission test | NFR-001, NFR-002 |
| **Reliability / concurrency** | Parallel duplicate submissions; clock-manipulation tests against server time | NFR-002, NFR-003 |
| **Usability and accessibility** | Viewport checks at 375/768/1280 px; axe DevTools and screen-reader check of form labels and errors | NFR-004, NFR-005 |
| **CI / pipeline** | Deliberately failing and passing commits on test branches to verify each pipeline gate | FR-027 to FR-031, NFR-007 |
| **Documentation** | Fresh-machine setup by a team member who did not write the README | NFR-006 |

## 5.3 Security validation

Planned security tests verify security objectives SO-01 and SO-02 and requirements SEC-001 to SEC-006. They use the OWASP Top 10 and the OWASP Web Security Testing Guide as a checklist. gitleaks and OWASP ZAP are planned supplementary tools; their scans will be run during the November test window and their reports attached to the Test Summary Report.

| Requirement | Validation technique | Test cases |
|---|---|---|
| **SEC-001 Password hashing** | Inspect stored user documents; confirm bcrypt hashes, unique salts, no plaintext | TC-SEC-01 |
| **SEC-002 JWT and roles** | Missing, malformed, tampered, wrong-secret, alg "none" and expired tokens; wrong-role requests | TC-AUTH-07, TC-AUTH-08, TC-AUTH-09, TC-SEC-02, TC-SEC-03 |
| **SEC-003 Answer-key exposure** | Automated response-schema checks on every student-facing endpoint | TC-SCR-04 |
| **SEC-004 Input validation** | Malformed and oversized input, invalid IDs, NoSQL injection, stored XSS payloads | TC-SEC-04, TC-SEC-05, TC-SEC-06 |
| **SEC-005 Secrets** | Planned gitleaks scan of full history; .gitignore and .env.example review | TC-SEC-07 |
| **SEC-006 Dependency audit** | npm audit in every CI run with a documented failure policy (fail on high/critical) | TC-CI-08, TC-SEC-08 |
| **SO-01 Unauthorized access** | Cross-user data access, role bypass, planned passive OWASP ZAP baseline scan | TC-SCR-06, TC-ADM-04, TC-SEC-10 |
| **SO-02 Fairness and integrity** | Repeat and concurrent attempts, late answers, clock tampering, client-supplied score, tie-break stability | TC-ATT-02, TC-ATT-11, TC-ATT-13, TC-SEC-09, TC-LB-03 |

Any defect that exposes answer keys, bypasses authorization or alters a finalized score is classified as Critical (Section 5.5).

## 5.4 Entry and exit criteria

| Phase | Entry criteria | Exit criteria |
|---|---|---|
| **Unit and integration** | Feature code on a feature branch; test DB configuration available | All tests pass in CI; frontend and backend statement coverage each at least 80% |
| **System** | SRS v1.0 baselined; build passes CI and smoke test; test data seeded; test cases reviewed | 100% of planned system TCs executed; all High-priority TCs pass |
| **Performance and security** | System test of the affected flows complete; tools installed | TC-PERF and TC-SEC results recorded; p95 within 2 s; no open Critical/High security defect |
| **Acceptance** | System test exit criteria met; release candidate tagged on main with a green CI run | AT-01 to AT-10 pass; faculty guide sign-off |

**Overall exit criteria:** 100% of planned test cases executed; 100% of High-priority test cases pass; 0 open Critical or High defects; all Medium/Low open defects documented with a workaround; requirement coverage 100%; latest CI run on main passes; Test Summary Report approved.

## 5.5 Defect management

Defects are logged as GitHub Issues with labels for severity and module, linked to the failing TC ID and the commit tested. Lifecycle: New, Assigned, Fixed, Ready for Retest, Closed (or Reopened if the retest fails). A fix is closed only after the TC passes and the regression suite is green.

| Severity | Definition | Example | Target resolution |
|---|---|---|---|
| **Critical** | Security breach, data loss, wrong score, or a core flow blocked | Answer keys returned to students; cannot submit a quiz | Before the next build |
| **High** | Major feature wrong with no reasonable workaround | Leaderboard tie-break wrong; second attempt allowed | Within 1 day |
| **Medium** | Feature partly wrong; workaround exists | Validation message missing on one field | Before submission if time allows |
| **Low** | Cosmetic or wording issue | Misaligned button at one width | Backlog |

## 6. Test Environment and Tools

| Component | Specification |
|---|---|
| **Client devices** | Team laptops (Windows/macOS) and one Android or iOS phone; Chrome DevTools device mode for 375 px, 768 px and 1280 px |
| **Browsers** | Latest stable Chrome, Firefox, Edge and Safari |
| **Application server** | Node.js LTS running the Express API locally with NODE_ENV=test |
| **Database** | Separate MongoDB test database (mongodb-memory-server for automated tests; local MongoDB or a dedicated Atlas test cluster for system tests). Production data is never used. |
| **CI environment** | GitHub Actions, ubuntu-latest runner, Node.js LTS, triggered on push and pull_request |
| **Configuration** | .env.test with a test-only JWT secret and short token expiry; secrets stored as GitHub Actions secrets, never committed |
| **Test data** | Seed script creating the users, quizzes and attempts listed in Appendix A; database reset before each automated run |

All tools below are planned for the November 2026 test window. Core tools are required by SRS Section 6.1 and the course CI/CD guidelines; supplementary tools add evidence for performance and security and are kept as planned tests.

| Tool | Purpose | Status |
|---|---|---|
| **Vitest + React Testing Library** | Unit and component tests; coverage reports | Planned - core (SRS 6.1) |
| **Supertest** | API integration tests against the Express app | Planned - core (SRS 6.1) |
| **ESLint, Prettier** | Static analysis and format checks in CI | Planned - core (SRS 6.1) |
| **npm audit** | Dependency security scan in CI | Planned - core (SRS 6.1) |
| **GitHub Actions** | CI execution, quality gates and report artifacts | Planned - core (SRS 6.2) |
| **k6** | Load test for NFR-001 and concurrent submission for NFR-002 | Planned - supplementary |
| **gitleaks** | Secret scan of the repository history (SEC-005) | Planned - supplementary |
| **OWASP ZAP (baseline scan)** | Passive web security scan of the running build | Planned - supplementary |
| **Postman** | Exploratory and manual API testing | Planned - supplementary |
| **Chrome DevTools, Lighthouse, axe DevTools** | Responsive, accessibility and form-error checks | Planned - supplementary |
| **GitHub Issues** | Defect tracking | Planned - supplementary |

## 7. Test Schedule and Milestones

Testing is scheduled for November 2026, as per the project timeline. Test planning and test case review are completed in October so that execution can start on the first working day of November. All dates are planned; changes will be recorded in the revision history.

| Milestone / activity | Start | End | Owner |
|---|---|---|---|
| **STP v1.0 prepared and shared with the team** | 12 Oct 2026 | 12 Oct 2026 | Samvith |
| **Test case review against SRS and SAD** | 13 Oct 2026 | 16 Oct 2026 | All |
| **Test environment, seed data and CI workflow ready** | 2 Nov 2026 | 4 Nov 2026 | Avyukth, Samvith |
| **Unit and integration testing (in CI, per feature branch)** | 2 Nov 2026 | 13 Nov 2026 | All developers |
| **System testing** | 16 Nov 2026 | 18 Nov 2026 | Samvith, Nainika |
| **Performance and security testing** | 19 Nov 2026 | 20 Nov 2026 | Samvith, Avyukth |
| **Defect fixing and regression testing** | 23 Nov 2026 | 25 Nov 2026 | All |
| **Acceptance testing (AT-01 to AT-10) / demonstration** | 26 Nov 2026 | 27 Nov 2026 | All, faculty guide |
| **Test Summary Report** | 30 Nov 2026 | 30 Nov 2026 | Samvith |

## 8. Test Deliverables

| Deliverable | Description | When |
|---|---|---|
| **Software Test Plan** | This document | Before testing |
| **Test case specification and RTM** | Section 13; kept in docs/testing/ and updated with each SRS change | Before testing |
| **Automated test scripts** | client and server test files; each test title starts with its TC ID | During development |
| **Test data and seed scripts** | Appendix A data set, .env.example entries for the test environment | Before system testing |
| **CI quality reports** | Test, coverage, lint and dependency-audit reports as downloadable artifacts from every GitHub Actions run (FR-030, course CI/CD guidelines) | Every run |
| **Performance and security reports** | Outputs of the planned k6, gitleaks and OWASP ZAP runs | After performance/security testing |
| **Test execution log** | TC ID, build/commit SHA, date, tester, result, defect link | During execution |
| **Defect reports** | GitHub Issues with severity, steps to reproduce and evidence | During execution |
| **Test Summary Report** | Results by module, coverage, open defects, deviations and release recommendation | End of testing |

## 9. Roles and Responsibilities

All three members develop across frontend, backend and database (SRS Appendix B), so every member writes unit and integration tests for their own code. The roles below define who owns each part of testing.

| Role | Name | Responsibility |
|---|---|---|
| **QA lead / test plan owner** | Samvith Vaddiparthi | Prepare and maintain the STP, test cases and RTM; set up test data; coordinate execution; run security and acceptance tests; publish the Test Summary Report |
| **Requirements owner / test engineer** | Batchu Hari Nainika | Owner of the SRS; clarify requirement intent; review test cases for coverage; execute system tests for authoring, feedback and administration |
| **Architecture owner / test engineer** | Avyukth KV | Owner of the SAD; review API-level test design; set up CI workflow and test database; run performance tests |
| **Developers** | All team members | Write unit and integration tests with each feature; keep CI green; fix and verify defects |
| **Product owner / approver** | Arpitha K (Faculty guide) | Review the plan and acceptance results; sign off readiness |

## 10. Risks and Mitigation

| Risk | Impact | Mitigation |
|---|---|---|
| **Tight submission timeline delays a stable build** | High | Risk-based order: run High-priority TCs first; smoke test on every build; automate API tests early |
| **Requirement changes after test design** | Medium | Version the SRS and STP together; update the RTM in the same pull request as the change |
| **Timer and concurrency defects are hard to reproduce** | High | Mock the server clock in unit tests; use short-duration test quizzes; parallel request tests (TC-ATT-07, TC-PERF-02) |
| **Shared or cloud test database unavailable** | Medium | Use mongodb-memory-server for automated tests so CI does not depend on Atlas |
| **Flaky tests in CI** | Medium | Reset and seed the database per run; no dependency on test order; fix or quarantine flaky tests within one day |
| **Secrets or test data leak into the repository** | High | Separate .env.test, GitHub Actions secrets, gitleaks scan (TC-SEC-07) |
| **Coverage gate (80%) not met near the deadline** | Medium | Write tests with each feature; check coverage on every pull request |
| **Limited hardware for load testing** | Low | Run k6 against a local build, record machine specs, and report results as indicative |
| **Team member unavailable** | Medium | Test cases and scripts stored in the repository so any member can execute them |

## 11. Assumptions and Dependencies

- SRS v1.0 (12 Oct 2026) is the requirements baseline; changes are reflected in this plan through the revision history.

- Endpoint names and component boundaries come from SAD v1.0; test cases describe behaviour, so they stay valid if routes are renamed.
- Score is the number of correct responses (SRS FR-019); unanswered questions score zero and there is no negative marking unless the SRS is changed.
- Each student has at most one attempt record per quiz (SRS FR-013), so each leaderboard entry corresponds to one student's attempt.
- Expiry may be enforced on the next request or by a scheduled job; tests check the outcome (finalized attempt, only answers saved before the deadline scored), not the mechanism.
- Seed data, a test MongoDB instance and GitHub Actions minutes are available before system testing starts.
- All team members are available for the acceptance demonstration.

# 12. Suspension and Resumption Criteria

## 12.1 Suspension criteria

- The build fails the smoke test (login, start quiz, submit, view score).
- Test environment or test database unavailable for more than 4 hours.
- Blocking defects prevent execution of more than 30% of the planned test cases in a cycle.
- A Critical security defect is found (e.g. answer keys exposed): testing of the affected module stops until it is fixed.
- The CI pipeline on main is broken, so build results cannot be trusted.

## 12.2 Resumption criteria

- Blocking defects are fixed, verified and merged, and the smoke test passes on the new build.
- Environment restored and test data re-seeded.
- CI on main is green again.
- The suspended test suite is re-executed from its beginning, and affected regression tests are re-run.

# 13. Test Case Management and Traceability

## 13.1 Test case conventions

Test case IDs use the format TC-<MODULE>-<NN> (AUTH, NAV, QM, ATT, SCR, LB, FB, ADM, CI, PERF, UX, SEC). Each test case references at least one SRS requirement. Automated tests include the TC ID in the test title, for example: it('TC-ATT-02 blocks a second attempt'), so CI reports can be traced back to this plan.

| Field | Content recorded for each test case |
|---|---|
| **Test case ID** | Unique ID, e.g. TC-ATT-08 |
| **Requirement(s)** | SRS IDs verified (FR / NFR / SEC / SO) |
| **Preconditions and test data** | Logged-in role, seeded quiz or attempt (Appendix A) |
| **Steps** | Actions performed in the UI or API calls made |
| **Expected result** | Observable outcome derived from the SRS acceptance criterion |
| **Level / priority** | U, I, S, A or P / H, M or L |
| **Execution record** | Build SHA, date, tester, actual result, Pass/Fail, defect link |

## 13.2 Test case specification

79 planned test cases cover all 46 trace items in SRS v1.0. Preconditions use the data set in Appendix A. Every test case currently has the status Planned; execution results (Pass/Fail, build SHA, date, tester) will be added to the test execution log during the November 2026 test window and summarized in the Test Summary Report.

### 13.2.1 Authentication and access control (TC-AUTH)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-AUTH-01** | FR-001 | Register a new student with a valid name, unused email and valid password. | Account is created with the Student role; success message shown; the user can log in. | I, S | H |
| **TC-AUTH-02** | FR-001 | Register again using an email that is already registered. | Registration rejected with a clear error; no second account is created. | I, S | H |
| **TC-AUTH-03** | FR-001<br>NFR-005 | Submit the registration form with an empty name, a malformed email and an empty password. | Field-level validation messages are shown; no account is created. | U, S | M |
| **TC-AUTH-04** | FR-002 | Log in with correct credentials for a student, a faculty member and an administrator. | Login succeeds, a token/session is issued and each user lands on the dashboard for their role. | I, S | H |
| **TC-AUTH-05** | FR-002 | Log in with a wrong password, then with an unregistered email. | Login fails with an error message; the response does not contain the password or password hash. | I, S | H |
| **TC-AUTH-06** | FR-003 | Log in and keep a copy of the issued token; log out; then call a protected API with that same token and open a protected page. | The logged-out token is rejected with 401 Unauthorized; the protected page redirects to Login. | I, S | H |
| **TC-AUTH-07** | FR-003<br>SEC-002 | Call protected endpoints with no Authorization header and with a malformed bearer token. | Each request returns 401 Unauthorized; no data is returned. | I | H |
| **TC-AUTH-08** | FR-004<br>SEC-002 | Using a student token, call quiz create, edit, publish and delete operations. | Each request returns 403 Forbidden; database is unchanged. | I, A | H |
| **TC-AUTH-09** | FR-004 | Using a faculty token, call the administrator role-management operation. | Request returns 403 Forbidden; the role is unchanged. | I | H |
| **TC-AUTH-10** | FR-004 | Log in as a student and look for Quiz Management in navigation; then open its URL directly. | Menu item is not shown; direct navigation is redirected or denied, and no management data is loaded. | S | M |

### 13.2.2 Quiz discovery and navigation (TC-NAV)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-NAV-01** | FR-005 | Seed published and draft quizzes in two topics; open Quiz Topics as a student. | Only published quizzes are listed, grouped under the correct topic. | I, S | H |
| **TC-NAV-02** | FR-005<br>FR-011 | As a student, request a draft quiz directly by its ID through the API. | Draft quiz is not returned to the student. | I | H |
| **TC-NAV-03** | FR-006 | Open the detail page of a published quiz. | Title, topic, duration and question count match the seeded quiz. | S | M |

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-NAV-04** | FR-007 | Use the navigation bar to open Home, About, Contact and Feedback. | Each page is reachable from navigation and renders without console or API errors. | S | M |
| **TC-NAV-05** | FR-007<br>NFR-004 | Repeat TC-NAV-04 at a 375 px viewport width. | Pages render with no horizontal scrolling; navigation remains usable. | S, A | M |

### 13.2.3 Quiz authoring and management (TC-QM)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-QM-01** | FR-008 | As faculty, create a quiz with title, topic, duration and questions. | Quiz is saved with Draft status and does not appear in student discovery. | I, S | H |
| **TC-QM-02** | FR-008 | Create a quiz with a missing title and a zero/negative duration. | Request rejected with validation errors; nothing is saved. | U, I | M |
| **TC-QM-03** | FR-009 | Add a multiple-choice question with four options and one correct option. | Question is saved and the correct option is stored with it. | I | H |
| **TC-QM-04** | FR-009 | Add a question with only one answer option. | Rejected: at least two options are required. | U, I | M |
| **TC-QM-05** | FR-009 | Add a question without selecting a correct option. | Rejected with a validation error; question is not saved. | U, I | M |
| **TC-QM-06** | FR-010 | As the owning faculty, edit a draft quiz title and question, then delete another own draft. | Edits persist after reload; deleted quiz no longer appears in Quiz Management. | I, S | H |
| **TC-QM-07** | FR-010<br>FR-004 | As faculty B, edit and delete a quiz owned by faculty A. | Both requests rejected (403); faculty A's quiz is unchanged. | I | H |
| **TC-QM-08** | FR-010 | As administrator, edit a draft quiz owned by a faculty member. | Edit succeeds and persists (administrators manage quizzes). | I | M |
| **TC-QM-09** | FR-011 | Publish a complete draft quiz, check student view, then unpublish it. | Quiz appears in student discovery after publishing and disappears after unpublishing. | I, S | H |
| **TC-QM-10** | FR-012 | Try to publish a quiz that has no questions, and one with a missing duration. | Publication rejected (422) with a message naming the missing fields; quiz stays Draft. | I, A | H |

### 13.2.4 Quiz attempt lifecycle (TC-ATT)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-ATT-01** | FR-013 | Student with no previous attempt starts a published quiz. | Attempt is created; server records start time and deadline; questions are returned. | I, S | H |
| **TC-ATT-02** | FR-013<br>SO-02 | After finalizing an attempt, try to start the same quiz again from the UI and directly through the API. | Second start is blocked by the API (409 Conflict); the first attempt and score are unchanged. | I, A | H |
| **TC-ATT-03** | FR-013 | Through the API, try to start a draft (unpublished) quiz. | Request rejected; no attempt is created. | I | M |
| **TC-ATT-04** | FR-014 | During an attempt, select answers and move between questions. | Options are selectable, selections stay visible when navigating, and the countdown is visible and decreasing. | S | H |
| **TC-ATT-05** | FR-015 | Select answers, then refresh the page during the active attempt. | Saved answers are restored; remaining time continues from the server deadline (not reset). | S | M |
| **TC-ATT-06** | FR-016 | Answer all questions and select Submit. | Attempt is recorded once as finalized and the score is returned; Submit cannot be used again. | I, S | H |
| **TC-ATT-07** | NFR-002<br>FR-016 | Send the submit request five times concurrently (and double-click Submit in the UI). | Exactly one finalized attempt is stored; repeated requests do not create duplicates or change the score. | I | H |
| **TC-ATT-08** | FR-017 | Start the 1-minute test quiz, save answers to two questions, then wait past the deadline without submitting. | Attempt is finalized automatically; only the answers saved before the deadline are scored. | I, A | H |
| **TC-ATT-09** | FR-018 | After submitting, send an answer update for the same attempt. | Update rejected; stored answers and score are unchanged. | I | H |
| **TC-ATT-10** | FR-018<br>FR-017 | After the deadline has passed (not submitted), send an answer update. | Update rejected; the finalized score is unchanged. | I, A | H |
| **TC-ATT-11** | NFR-003<br>SO-02 | Move the browser/client clock back and edit timer values in browser storage, then keep answering past the real deadline. | Server still enforces the original deadline; late answers are rejected. | I, S | H |

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-ATT-12** | FR-017<br>NFR-003 | Unit-test deadline logic with a mocked server clock: save an answer 1 s before and 1 s after the deadline. | Save before the deadline is accepted; save after the deadline is rejected. | U | H |
| **TC-ATT-13** | FR-013<br>SO-02 | Send five start requests for the same student and quiz at the same moment. | Exactly one attempt record exists afterwards; the other requests return that same attempt or 409 Conflict. | I | H |
| **TC-ATT-14** | FR-013<br>FR-015 | With an active attempt still before its deadline, start the same quiz again (e.g. from a new tab). | The existing attempt is resumed with its saved answers and original deadline; no new attempt is created. | I, S | H |

### 13.2.5 Scoring, history and leaderboard (TC-SCR, TC-LB)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-SCR-01** | FR-019 | Unit-test the scoring function for a 5-question quiz with all correct, all wrong, 3 correct and all unanswered. | Scores are 5, 0, 3 and 0 respectively (score = number of correct responses). | U | H |
| **TC-SCR-02** | FR-019<br>SO-02 | Submit a known answer set for the seeded quiz through the API. | Stored score equals the expected value computed from the stored answer key. | I | H |
| **TC-SCR-03** | FR-020 | After submission, view the result page. | Score is displayed; correct answers and explanations are not displayed. | S, A | H |
| **TC-SCR-04** | FR-020<br>SEC-003 | Inspect JSON returned to a student by quiz detail, start attempt, save answer, submit/result, My Scores and leaderboard endpoints. | No correct-option, answer-key or explanation fields are present in any response. | I, A | H |
| **TC-SCR-05** | FR-021 | Student with two finalized attempts opens My Scores. | Both attempts are listed with quiz title, score and submission date; no other student's attempts appear. | I, S | M |
| **TC-SCR-06** | FR-021<br>SO-01 | As student A, request student B's score history by changing the identifier in the request. | Request denied or only student A's own data is returned. | I | H |
| **TC-LB-01** | FR-022 | Create finalized attempts for two different quizzes; open each quiz's leaderboard. | Each leaderboard contains only attempts for its own quiz. | I, A | H |
| **TC-LB-02** | FR-023 | Seed attempts with scores 4, 2 and 5 for one quiz; open its leaderboard. | Entries are ordered 5, 4, 2 (highest score first). | U, I | H |
| **TC-LB-03** | FR-023<br>SO-02 | Seed two attempts with equal scores and different submission timestamps; load the leaderboard several times. | Earlier submission ranks higher, and the order is identical on every load. | U, I | H |

### 13.2.6 Feedback and administration (TC-FB, TC-ADM)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-FB-01** | FR-024 | Logged-in user submits feedback with a valid rating and a message. | Feedback is stored and an acknowledgement is shown. | I, S | M |
| **TC-FB-02** | FR-024<br>NFR-005 | Submit feedback with neither a rating nor a message. | Validation error shown; nothing is stored. | U, I | M |
| **TC-FB-03** | FR-024<br>SEC-002 | Submit feedback through the API without a token. | Request returns 401 Unauthorized; nothing is stored. | I | M |
| **TC-ADM-01** | FR-025 | Administrator opens the feedback view. | All submitted feedback is listed. | I, S | M |
| **TC-ADM-02** | FR-025<br>FR-004 | Student opens the feedback view and calls the feedback-list API. | Page is not accessible and the API returns 403 Forbidden. | I, S | H |
| **TC-ADM-03** | FR-026 | Administrator changes a user's role from Student to Faculty. | Role change is saved and shown in the user list; the user receives Faculty permissions. | I, S | M |
| **TC-ADM-04** | FR-026<br>FR-004 | Faculty and student users attempt to change any user's role through the API. | Requests return 403 Forbidden; roles are unchanged. | I, A | H |

### 13.2.7 Version control and CI pipeline (TC-CI)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-CI-01** | FR-027 | Review repository history and pull requests. | Features were built on feature branches, merged through reviewed PRs, with descriptive commit messages. | S | M |
| **TC-CI-02** | FR-028 | Push a commit to a feature branch and open a pull request. | The GitHub Actions workflow starts automatically for both the push and the pull-request event. | S | H |

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-CI-03** | FR-029 | On a test branch, introduce an ESLint error and a Prettier formatting violation. | CI fails at the lint/format step. | S | H |
| **TC-CI-04** | FR-029 | Push a deliberately failing unit test to a feature branch. | CI run fails and the PR cannot be merged into main. | S, A | H |
| **TC-CI-05** | FR-029<br>NFR-007 | Push a correct implementation with passing tests. | Lint, format, tests, coverage (frontend and backend), audit and build steps all pass. | S, A | H |
| **TC-CI-06** | FR-030 | Open a completed workflow run. | Test and coverage reports (FR-030), plus lint and dependency-audit reports required by the course CI/CD guidelines, are downloadable artifacts. | S | M |
| **TC-CI-07** | FR-031<br>NFR-007 | Check that frontend and backend coverage are reported separately; then add an untested module so one of them falls below 80%. | Two coverage reports are produced; CI fails at the coverage gate unless an approved, documented exception exists. | S | H |
| **TC-CI-08** | FR-031<br>SEC-006 | On a throwaway branch, add a dependency version with a known high-severity advisory. | Dependency audit reports it and CI applies the documented failure policy (run fails). | S | M |
| **TC-CI-09** | FR-029 | Introduce a frontend import error that only breaks the production build. | CI fails at the build step. | S | M |

### 13.2.8 Performance, usability and maintainability (TC-PERF, TC-UX)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-PERF-01** | NFR-001 | Load test (planned, k6): 50 concurrent virtual users for 5 minutes running login, list quizzes, open quiz, save answer, submit and view leaderboard. | p95 response time is 2 s or less; error rate is recorded in the report. | P | M |
| **TC-PERF-02** | NFR-002<br>FR-016 | 50 students submit the same quiz at the same moment. | Every student has exactly one finalized attempt; no attempt is lost or duplicated. | P, I | H |
| **TC-UX-01** | NFR-004 | Open every core page at 375 px, 768 px and 1280 px widths. | No horizontal scrolling at any width; content and controls remain readable and usable. | S, A | M |
| **TC-UX-02** | NFR-005 | Submit Register, Login, Quiz create and Feedback forms with required fields empty; check with axe DevTools and a screen reader. | Every field has a visible label; errors explain how to fix the input and are announced to assistive technology. | S | M |
| **TC-UX-03** | NFR-006 | A team member follows README on a clean machine: clone, copy .env.example, install, seed, run app and tests; review API documentation. | Project runs with no undocumented steps; API documentation covers the implemented endpoints. | S | M |

### 13.2.9 Security validation (TC-SEC)

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-SEC-01** | SEC-001<br>SO-01 | Register two users with the same password; inspect both user records in the test database. | Passwords are stored as bcrypt hashes (never plaintext) and the two hashes differ (salted). | I | H |
| **TC-SEC-02** | SEC-002<br>SO-01 | Call protected endpoints with a token whose payload was edited (role set to admin), one signed with the wrong secret, and one with alg "none". | All requests return 401 Unauthorized. | I | H |
| **TC-SEC-03** | SEC-002 | Call a protected endpoint with an expired token (short expiry set in .env.test). | Request returns 401 Unauthorized. | I | H |
| **TC-SEC-04** | SEC-004 | Send malformed JSON, wrong data types, very long strings and invalid IDs to each write endpoint. | Requests fail with 4xx validation errors; server keeps running; no stack trace in responses. | I | H |
| **TC-SEC-05** | SEC-004<br>SO-01 | Attempt NoSQL injection on login, e.g. email and password sent as {"$ne": null}. | Request rejected; no session is issued. | I | H |
| **TC-SEC-06** | SEC-004 | Save <script>alert(1)</script> as a quiz title and as feedback text; open the pages that display them. | Content is shown as plain text; the script does not run. | S | H |
| **TC-SEC-07** | SEC-005 | Run the planned gitleaks scan over the full git history; check .gitignore and .env.example. | No secrets found; .env is ignored; .env.example holds placeholders only. | S | H |
| **TC-SEC-08** | SEC-006 | Review CI logs of recent runs and the README failure policy. | Dependency audit runs on every CI run and the documented failure policy is applied. | S | M |

| TC ID | Req. | Scenario / steps | Expected result | Level | Pri. |
|---|---|---|---|---|---|
| **TC-SEC-09** | SO-02<br>FR-019 | On submit, include extra fields such as score: 100 or a modified answer key in the request body. | Server ignores client-supplied score/key fields and computes the score itself. | I | H |
| **TC-SEC-10** | SO-01<br>SEC-004 | Run the planned OWASP ZAP baseline (passive) scan against the locally running build. | No High-risk alerts; Medium alerts are triaged and recorded in the test summary. | S | M |

## 13.3 Requirements Traceability Matrix (RTM)

Every SRS requirement maps to at least one planned test case. The SRS acceptance column repeats the SRS Section 8 mapping so the two documents stay consistent.

| Req. ID | Requirement (summary) | Test cases | SRS acceptance |
|---|---|---|---|
| **FR-001** | Student registration with name, email and password | TC-AUTH-01, TC-AUTH-02, TC-AUTH-03 | AT-01, AT-06 |
| **FR-002** | Authentication and authorized session | TC-AUTH-04, TC-AUTH-05 | AT-01, AT-06 |
| **FR-003** | Logout invalidates the session/token for later protected requests | TC-AUTH-06, TC-AUTH-07 | AT-01, AT-06 |
| **FR-004** | Server-side student / faculty / administrator permissions | TC-AUTH-08, TC-AUTH-09, TC-AUTH-10, TC-QM-07, TC-ADM-02, TC-ADM-04 | AT-01, AT-06 |
| **FR-005** | Published quizzes displayed grouped by topic | TC-NAV-01, TC-NAV-02 | AT-10 |
| **FR-006** | Quiz title, topic, duration and question count shown before attempt | TC-NAV-03 | AT-10 |
| **FR-007** | Home, About, Contact and Feedback pages | TC-NAV-04, TC-NAV-05 | AT-10 |
| **FR-008** | Faculty/admin create quiz (title, topic, duration, questions) | TC-QM-01, TC-QM-02 | AT-07 |
| **FR-009** | Multiple-choice questions with options and stored correct option | TC-QM-03, TC-QM-04, TC-QM-05 | AT-07 |
| **FR-010** | Edit and delete authorized draft quizzes | TC-QM-06, TC-QM-07, TC-QM-08 | AT-07 |
| **FR-011** | Publish / unpublish quiz | TC-NAV-02, TC-QM-09 | AT-07 |
| **FR-012** | Required quiz fields validated before publication | TC-QM-10 | AT-07 |
| **FR-013** | One attempt record per quiz; active attempt resumable, finalized attempt not restartable | TC-ATT-01, TC-ATT-02, TC-ATT-03, TC-ATT-13, TC-ATT-14 | AT-02, AT-03 |
| **FR-014** | Questions with selectable answers and visible countdown | TC-ATT-04 | AT-02, AT-03 |
| **FR-015** | Selected answers preserved during active attempt | TC-ATT-05, TC-ATT-14 | AT-02, AT-03 |
| **FR-016** | Attempt submitted when student selects Submit | TC-ATT-06, TC-ATT-07, TC-PERF-02 | AT-02, AT-03 |
| **FR-017** | Automatic finalization at server-enforced time limit | TC-ATT-08, TC-ATT-10, TC-ATT-12 | AT-02, AT-03 |
| **FR-018** | Answer updates rejected after submission or expiry | TC-ATT-09, TC-ATT-10 | AT-02, AT-03 |
| **FR-019** | Automatic scoring using stored answer keys | TC-SCR-01, TC-SCR-02, TC-SEC-09 | AT-04, AT-05 |
| **FR-020** | Score shown without correct answers or explanations | TC-SCR-03, TC-SCR-04 | AT-04, AT-05 |
| **FR-021** | History of own completed attempts (My Scores) | TC-SCR-05, TC-SCR-06 | AT-04, AT-05 |
| **FR-022** | Separate leaderboard for each quiz | TC-LB-01 | AT-04, AT-05 |
| **FR-023** | Ranking by score with earlier-submission tie-break | TC-LB-02, TC-LB-03 | AT-04, AT-05 |
| **FR-024** | Authenticated users submit feedback (rating or message) | TC-FB-01, TC-FB-02, TC-FB-03 | AT-06 |
| **FR-025** | Administrators view submitted feedback | TC-ADM-01, TC-ADM-02 | AT-06 |
| **FR-026** | Administrators manage user roles | TC-ADM-03, TC-ADM-04 | AT-06 |
| **FR-027** | Feature branches and pull-request review | TC-CI-01 | AT-08, AT-09 |
| **FR-028** | CI runs automatically on pushes and pull requests | TC-CI-02 | AT-08, AT-09 |
| **FR-029** | CI runs lint, format check, tests and production build | TC-CI-03, TC-CI-04, TC-CI-05, TC-CI-09 | AT-08, AT-09 |
| **FR-030** | CI publishes test and coverage reports as artifacts | TC-CI-06 | AT-08, AT-09 |
| **FR-031** | CI coverage gate and dependency security checks | TC-CI-07, TC-CI-08 | AT-08, AT-09 |
| **NFR-001** | p95 API response time ≤ 2 s at 50 concurrent users | TC-PERF-01 | Load / concurrency |
| **NFR-002** | Idempotent quiz finalization | TC-ATT-07, TC-PERF-02 | Load / concurrency |
| **NFR-003** | Quiz expiry validated against server time | TC-ATT-11, TC-ATT-12 | Load / concurrency |
| **NFR-004** | No horizontal scrolling at widths ≥ 375 px | TC-NAV-05, TC-UX-01 | AT-09, AT-10 |
| **NFR-005** | Form labels and actionable validation messages | TC-AUTH-03, TC-FB-02, TC-UX-02 | AT-09, AT-10 |
| **NFR-006** | Setup instructions, .env example and API documentation | TC-UX-03 | AT-09, AT-10 |
| **NFR-007** | Statement coverage ≥ 80%, reported separately for frontend and backend | TC-CI-05, TC-CI-07 | AT-09, AT-10 |
| **SEC-001** | Passwords stored as salted cryptographic hashes | TC-SEC-01 | AT-04, AT-06 |
| **SEC-002** | JWT signature and role validation on protected endpoints | TC-AUTH-07, TC-AUTH-08, TC-FB-03, TC-SEC-02, TC-SEC-03 | AT-04, AT-06 |

| Req. ID | Requirement (summary) | Test cases | SRS acceptance |
|---|---|---|---|
| **SEC-003** | No answer keys in student-facing responses | TC-SCR-04 | AT-04, AT-06 |
| **SEC-004** | Validation and sanitization of user-controlled input | TC-SEC-04, TC-SEC-05, TC-SEC-06, TC-SEC-10 | AT-04, AT-06 |
| **SEC-005** | Secrets loaded from environment variables | TC-SEC-07 | AT-04, AT-06 |
| **SEC-006** | Dependency security auditing in CI | TC-CI-08, TC-SEC-08 | AT-04, AT-06 |
| **SO-01** | Protect accounts, answer keys and admin operations | TC-SCR-06, TC-SEC-01, TC-SEC-02, TC-SEC-05, TC-SEC-10 | Security tests |
| **SO-02** | Integrity and fairness of attempts, deadlines, scores and leaderboards | TC-ATT-02, TC-ATT-11, TC-ATT-13, TC-SCR-02, TC-LB-03, TC-SEC-09 | Security tests |

### 13.4 Acceptance test mapping

| AT ID | Scenario (SRS) | Expected outcome (SRS) | Verified by |
|---|---|---|---|
| **AT-01** | Register, login, logout | Valid user can access dashboard; invalid credentials rejected | TC-AUTH-01, TC-AUTH-04, TC-AUTH-05, TC-AUTH-06 |
| **AT-02** | Student attempts quiz twice | First finalized attempt saved; second start denied | TC-ATT-01, TC-ATT-02, TC-ATT-13 |
| **AT-03** | Quiz timer expires | Attempt finalized with saved answers only | TC-ATT-08, TC-ATT-10 |
| **AT-04** | Student checks result | Score shown; correct answers hidden | TC-SCR-03, TC-SCR-04 |
| **AT-05** | Open two quiz leaderboards | Each shows only its own quiz attempts | TC-LB-01 |
| **AT-06** | Student calls admin API | 403 Forbidden response | TC-AUTH-08, TC-ADM-02, TC-ADM-04 |
| **AT-07** | Faculty publishes incomplete quiz | Validation rejects publication | TC-QM-10 |
| **AT-08** | Push failing test to feature branch | CI run fails | TC-CI-04 |
| **AT-09** | Push passing implementation | Lint, test, coverage, audit and build pass | TC-CI-05 |
| **AT-10** | Inspect responsive UI | No horizontal overflow at 375 px | TC-NAV-05, TC-UX-01 |

## 14. Test Metrics and Reporting

| Metric | Calculation | Target |
|---|---|---|
| **Test execution progress** | Executed TCs / planned TCs × 100 | 100% at exit |
| **Pass rate** | Passed TCs / executed TCs × 100 | 100% of High priority; at least 95% overall |
| **Requirement coverage** | Requirements with at least one passed TC / total requirements × 100 | 100% |
| **Statement coverage** | From the frontend and backend CI coverage reports | Each at least 80% (NFR-007) |
| **Defect density** | Defects found per module (TI-01 to TI-09) | Tracked to target regression testing |
| **Defect aging** | Days from New to Closed, by severity | Critical: before next build; High: 1 day |
| **Open defects by severity** | Count of open Critical / High / Medium / Low | 0 Critical and 0 High at exit |
| **API p95 response time** | k6 result at 50 virtual users | 2 s or less (NFR-001) |
| **CI health** | Result of the latest workflow run on main | Passing |

**Reports:** daily status update to the team (executed, passed, failed, blocked, new defects); CI test and coverage artifacts on every run; defect reports in GitHub Issues; and a final Test Summary Report with results per module, metrics against targets, open defects with workarounds, deviations from this plan and a release recommendation.

## 15. Approvals

| Role | Name | Signature / Date |
|---|---|---|
| **Prepared by - QA lead** | Samvith Vaddiparthi | Pending |
| **Reviewed by - Requirements owner** | Batchu Hari Nainika | Pending |
| **Reviewed by - Architecture owner** | Avyukth KV | Pending |
| **Approved by - Faculty guide** | Arpitha K | Pending |

## Appendix A. Test Data Set

Created by the seed script in the test database only. Passwords for test accounts are defined in the seed configuration (.env.test) and are not written in this document.

| Data | Values | Used by |
|---|---|---|
| **Users** | admin01 (Administrator); faculty01, faculty02 (Faculty); student01 to student05 (Student); emails in the reserved example.com domain | All modules |
| **Quiz Q1 'JavaScript Basics'** | Topic Programming; published; 10 min; 5 questions; owner faculty01 | NAV, ATT, SCR, LB, PERF |
| **Quiz Q2 'Data Structures'** | Topic CS Fundamentals; published; 15 min; 5 questions; owner faculty02 | NAV, LB-01, QM-07 |
| **Quiz Q3 'Networks' (draft)** | Complete draft, not published; owner faculty01 | NAV-01, NAV-02, ATT-03, QM-09 |
| **Quiz Q4 (incomplete draft)** | Draft with no questions and missing duration | QM-10 (AT-07) |
| **Quiz Q5 'Timer Test'** | Published; 1 min; 3 questions | ATT-08, ATT-10, ATT-11 |
| **Leaderboard attempts** | Q1 attempts scoring 4, 2 and 5; two equal scores with submission times 1 minute apart | LB-02, LB-03 |
| **Tokens** | Expired token; tampered-payload token; token signed with a wrong secret; alg "none" token | AUTH-07, SEC-02, SEC-03 |
| **Malicious inputs** | {"$ne": null} objects, <script>alert(1)</script>, 10,000-character strings, invalid IDs | SEC-04, SEC-05, SEC-06 |